```
> #TASK 3.1.2
> with(numtheory) :
> with(RandomTools) :
> p := nextprime(Generate(integer(range=2..10^4)));
                              p := 7933                                   (1)
> a := nextprime(Generate(integer(range=2..10^4)));
                              a := 8059                                   (2)
> igcd(a, p − 1);
                                  1                                       (3)
> alpha := a&^(phi(phi(p)) − 1) mod phi(p);
                              α := 5059                                   (4)
> b := nextprime(Generate(integer(range=2..10^4)));
                                b := 7                                    (5)
> igcd(b, p − 1);
                                  1                                       (6)
> beta := b&^(phi(phi(p)) − 1) mod phi(p);
                              β := 6799                                   (7)
> m := Generate(integer(range=10^5..10^15)) mod (p − 1) #m<(p-1)
                              m := 6652                                   (8)
> m1 := m&^a mod p;
                             m1 := 1339                                   (9)
> m2 := m1&^b mod p;
                              m2 := 668                                   (10)
> m3 := m2&^alpha mod p;
                             m3 := 6353                                   (11)
> m4 := m3&^beta mod p;
                             m4 := 6652                                   (12)
> m − m4; #CHECK
                                  0                                       (13)
>
```